

SASKATOON KERNEN FARM, SK, Canada

WMO: 715131

Lat: 52.150N

Lon: 106.550W

Elev: 1673

StdP: 13.83

Time Zone: -6.00 (W06)

Period: 96-05

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.3	-23.0	-35.4	0.8	-28.3	-29.3	1.1	-23.0	24.8	7.4	22.4	9.5	6.6	150	0.718

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.6	87.2	63.8	83.4	62.4	80.2	61.0	68.9	80.5	66.5	76.9	64.4	74.5	11.1	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.1	98.8	74.6	62.6	90.3	71.6	60.3	83.2	69.7	34.1	80.6	32.1	76.7	30.3	74.7	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.0	21.2	19.0	DB	-34.4	95.6	3.5	4.0	-36.9	98.4	-39.0	100.8	-40.9	103.0	-43.5	106.0	(4)
(5)			WB	-36.2	72.6	5.7	2.9	-40.3	74.7	-43.6	76.4	-46.8	78.0	-51.0	80.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.5	1.9	11.7	20.4	39.2	50.2	59.2	65.0	63.7	53.2	38.2	23.2	10.9		
	DBStd	23.98	15.15	12.52	14.62	11.11	8.44	6.88	6.07	7.34	8.09	9.46	11.80	14.59		
	HDD50	6391	1491	1074	919	347	103	6	0	1	57	378	803	1212		
	HDD65	10581	1956	1494	1384	774	462	200	75	115	360	831	1253	1677		
	CDD50	1471	0	0	0	24	109	282	464	426	154	12	0	0		
	CDD65	184	0	0	0	0	3	25	74	75	7	0	0	0		
	CDH74	2599	0	0	0	23	96	388	926	1001	160	5	0	0		
	CDH80	781	0	0	0	7	9	97	277	361	30	0	0	0		
	WSAvg	9.3	8.9	8.8	9.7	10.8	11.0	9.5	8.5	8.2	9.2	9.3	8.7	9.0		
	PrecAvg	14.3	0.7	0.6	0.7	0.8	1.8	2.6	2.3	1.5	1.3	0.7	0.6	0.7		
	PrecMax	19.1	1.7	1.3	1.9	2.4	5.9	6.4	5.8	3.6	2.8	3.0	1.6	1.3		
(15) Precipitation	PrecMin	9.3	0.2	0.0	0.2	0.1	0.3	0.5	0.6	0.1	0.2	0.0	0.0	0.2		
	PrecStd	2.3	0.4	0.4	0.4	0.6	1.3	1.4	1.3	0.9	0.7	0.7	0.4	0.3		
	DB	38.0	39.5	52.5	76.4	81.2	89.2	91.2	94.2	84.0	73.1	53.6	42.8			
	MCWB	34.5	35.9	42.9	51.6	56.7	64.7	68.0	62.2	58.6	52.3	44.7	37.6			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	33.7	35.4	46.0	68.2	77.0	81.8	86.2	88.1	78.6	64.9	46.4	38.4			
	2% MCWB	31.5	32.4	38.8	48.4	54.1	59.4	65.5	61.8	56.9	49.1	40.5	34.6			
	5% DB	28.7	32.7	42.1	63.1	72.4	78.4	82.7	84.0	73.8	59.0	41.8	34.4			
	5% MCWB	26.8	30.5	36.9	47.6	52.5	59.1	63.8	62.9	55.3	47.8	37.4	32.0			
	10% DB	23.3	29.4	38.1	57.6	67.9	74.3	79.2	79.9	68.7	53.6	38.3	30.5			
	10% MCWB	21.7	27.4	34.3	45.4	50.7	57.9	63.4	61.9	53.6	44.4	34.9	28.7			
	0.4% WB	35.0	35.8	43.9	54.5	60.8	68.4	73.6	70.8	62.6	54.0	45.8	37.9			
	0.4% MCDB	37.1	38.9	50.6	70.2	74.3	80.3	85.7	81.1	76.7	69.4	52.8	41.8			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	31.2	32.9	40.1	51.5	56.3	64.5	70.1	67.9	59.4	51.1	41.1	34.7			
	2% MCDB	33.6	35.0	44.9	64.7	70.4	75.8	82.8	80.0	72.0	62.1	45.8	37.9			
	5% WB	26.9	30.6	37.0	48.9	54.0	62.1	67.6	65.6	57.3	48.1	37.9	32.0			
	5% MCDB	28.5	32.9	41.3	59.9	68.8	73.0	78.0	76.4	69.5	57.6	41.2	34.6			
	10% WB	21.7	27.5	34.4	46.2	51.9	59.8	65.5	63.5	55.2	45.2	35.1	28.8			
	10% MCDB	23.1	29.3	38.0	57.0	65.3	69.8	75.0	74.4	65.8	52.8	38.0	30.7			
	MDBR	17.7	18.4	18.3	21.1	24.3	21.0	22.6	24.3	23.1	20.7	16.8	17.2			
	5% DB	20.9	18.7	19.9	30.3	31.8	28.3	28.4	31.5	33.2	32.0	21.0	20.4			
(36) Mean Daily Temperature Range	5% WB	19.2	16.4	14.0	16.2	15.5	12.3	13.0	12.7	16.3	19.4	15.9	17.9			
	MCDBR	21.1	18.5	18.4	26.8	26.7	22.7	25.3	25.7	26.5	29.6	19.9	19.9			
	MCWBR	19.5	16.3	13.7	15.3	13.8	12.1	13.6	13.5	14.6	19.0	15.4	17.7			
	taub	0.251	0.266	0.297	0.340	0.354	0.374	0.374	0.376	0.322	0.282	0.253	0.219			
(40) Clear-Sky Solar Irradiance	taud	2.284	2.269	2.336	2.348	2.355	2.349	2.348	2.338	2.474	2.557	2.472	2.372			
	Ebn at Noon	243	273	286	285	285	279	276	269	275	269	247	244			
	Edh at Noon	20	27	31	35	37	37	37	35	27	20	16	15			
	RadAvg	351	681	1151	1519	1870	1891	1991	1615	1173	689	347	253			
(44) All-Sky Solar Radiation	RadStd	33	42	86	148	146	149	111	120	111	101	41	27			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page